

2007

一. 解

$$(1) (\varepsilon_1, \varepsilon_2, \varepsilon_3) = (\alpha_1, \alpha_2, \alpha_3)P \Rightarrow P = \begin{bmatrix} -1 & 1 & -1 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$B = P^{-1}AP$$

$$(2) f(\alpha_1, \alpha_2, \alpha_3) = (\alpha_1, \alpha_2, \alpha_3)A = (f(\alpha_1), f(\alpha_2), f(\alpha_3)) = (\alpha_1 + \alpha_2 - \alpha_3, \alpha_2 + 2\alpha_3, -\alpha_1 + 3\alpha_3)$$

$$\text{令 } X = (\alpha_1, \alpha_2, \alpha_3)^T, \text{ 求 } AX = 0$$

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ -1 & 2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \text{ 求得 } X = (1, -1, 1)^T$$

$$\therefore \text{核子空间 } N(f) \text{ 的基是 } \alpha_1, \alpha_1 + \alpha_2 + \alpha_3 + \alpha_3 = \alpha_1 - \alpha_2 + \alpha_3 = (-2, 2, 3)^T$$

$$f \text{ 的值域 } R(f) = \text{span}\{f(\alpha_1), f(\alpha_2), f(\alpha_3)\}$$

$$= \text{span}\{\alpha_1 + \alpha_2 - \alpha_3, \alpha_2 + 2\alpha_3, -\alpha_1 + 3\alpha_3\}$$

$$= \text{span}\{[0, 0, -1]^T, [1, 2, 1]^T\} \quad \text{找线性无关}$$

$$\text{二. } AA^H = \begin{bmatrix} 2 & i \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ -i & 0 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \therefore \text{奇异值为 } \sqrt{5}$$

$$\text{其解 } AA^H - 3E = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix} \sim \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \therefore (AA^H - 3E)X = 0 \text{ 的解向量为 } \alpha_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \therefore U_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$AA^H X = 0 \text{ 的解向量为 } \alpha_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \alpha_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \therefore U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$V_1 = A^H U_1 \Delta^{-1} = \begin{bmatrix} 2 & 0 & 0 \\ -i & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \frac{1}{\sqrt{5}} = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ -\frac{i}{\sqrt{5}} \end{bmatrix} \therefore V = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{i}{\sqrt{5}} \\ \frac{i}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix}$$

$$\therefore \text{奇异值分解式为 } A = U \begin{bmatrix} \sqrt{5} & 0 \\ 0 & 0 \end{bmatrix} V^H$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{5} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{i}{\sqrt{5}} \\ \frac{i}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix}$$

三. 解:

$$|A - \lambda E| = \begin{vmatrix} 1-\lambda & -1 & 1 \\ 2 & -2-\lambda & 2 \\ -1 & 1 & -1-\lambda \end{vmatrix} = -\lambda^2(\lambda+2)$$

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -2 & 2 \\ -1 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad Y(A - 0E) = 1, \therefore \lambda = 0 \text{ 所对应的两个线性无关的向量}$$

$$\therefore A \text{ 是对角化矩阵, } A = P \text{diag}(\lambda_1, \lambda_2, \lambda_3)P^{-1} = P \text{diag}(0, 0, -2)P^{-1}$$

$$\text{求 } (A - 0E)X = 0 \text{ 得: } \alpha_1 = (-1, 0, 1)^T, \alpha_2 = (1, 1, 0)^T$$

$$(A + 2E)X = 0 \text{ 得: } \alpha_3 = (-1, -2, 1)^T$$

$$\therefore P = (\alpha_1, \alpha_2, \alpha_3) = \begin{bmatrix} -1 & 1 & -1 \\ 0 & 1 & -2 \\ 1 & 0 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 1 & 0 & 1 \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \therefore \begin{aligned} \beta_1 &= (-\frac{1}{2}, \frac{1}{2}, \frac{1}{2})^T \\ \beta_2 &= (1, 0, 1)^T \\ \beta_3 &= (\frac{1}{2}, -\frac{1}{2}, \frac{1}{2})^T \end{aligned}$$

$$\therefore G_1 = \alpha_1 \beta_1^T + \alpha_2 \beta_2^T =$$

$$G_2 = \alpha_3 \beta_3^T$$

$$\text{则 } A = -2G_2$$

四. 证明: (1) $\|A\|_2 = \max(\lambda_i(A^H A))^{\frac{1}{2}}$

$$A = I - UU^T, A^H = I - UU^T = A$$

$$A^H A = A^2 = (I - UU^T)(I - UU^T) = I - 2UU^T + UU^T UU^T = I - UU^T = A$$

$\therefore A^H A$ 的特征值即为 A 的特征值

又 $A^2 = A \therefore P^{-1}AP = \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix} \therefore A$ 的特征值非 0 即 1 $\therefore \|A\|_2 = 1$

(2) 若 $Y = AX - AX$, 则 $(Y, AX) = 0$

$$\text{即: } Y^T(AX) = Y(AX)^T = 0$$

$$\therefore \|X\|_2 = \|AX + Y\|_2 = (AX + Y)^T(AX + Y) = \|AX\|_2 + \|Y\|_2$$

五. 同 2005 年 $\therefore \|AX\|_2 \leq \|X\|_2$

六. 证

(1) A 的最小多项式 $m(\lambda) = (\lambda - 3)^2$

$$\text{设 } p(x) = a_0 + a_1 x$$

$$\text{则 } p(3) = f(3) = a_0 + 3a_1, p'(3) = f'(3) = a_1$$

$$\therefore a_0 = f(3) - 3f'(3), a_1 = f'(3)$$

$$\text{则 } p(x) = [f(3) - 3f'(3)] + f'(3)x$$

$$f(A) = [f(3) - 3f'(3)]I + f'(3)A$$

$$\textcircled{1} f(x) = e^{tx}, f(3) = e^{3t}, f'(3) = te^{3t}$$

$$e^{tA} = [e^{3t} - 3te^{3t}]I + te^{3t}A$$

= ...

$$(2) \textcircled{2} A^T = \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix} \quad \varphi_{A^T}(\lambda) = (\lambda - 3)^4$$

$$\therefore A \sim A^T, \therefore \varphi_A(\lambda) = (\lambda - 3)^4$$

$$\text{设 } p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$$

...

$$(3) |A - \lambda E| = (2 - \lambda)(\lambda^2 - 1)$$

$$\text{其 Jordan 标准形为 } \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\varphi_A(\lambda) = (\lambda - 2)(\lambda - 1)(\lambda + 1)$$

$$\text{设 } p(x) = a_0 + a_1 x + a_2 x^2$$

七. 证: 求导: 两边对 t 求导

$$A \cos t A = \begin{bmatrix} 5 \cos 5t + 3 \cos t & 10 \cos 5t - 2 \cos t & 5 \cos 5t - \cos t \\ 5 \cos 5t - \cos t & 10 \cos 5t + 2 \cos t & 5 \cos 5t - \cos t \\ 5 \cos 5t - \cos t & 10 \cos 5t - 2 \cos t & 5 \cos 5t + 3 \cos t \end{bmatrix}$$

$$\text{令 } t = 0 \text{ 则 } A = \begin{bmatrix} 8 & 8 & 4 \\ 4 & 12 & 4 \\ 4 & 8 & 8 \end{bmatrix}$$

八. 证: 由已知 $AX = b$

$$\text{其中 } A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}, b = [1 \ 0 \ 1 \ 0]^T$$

则最佳最小二乘解为 $x = A^+ b$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \therefore B = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\therefore A^+ = C^H(C C^H)^{-1}(B^H B)^{-1} B^H =$$

九. 证:

$$A = \begin{bmatrix} 0 & -i & 1 \\ i & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$|\lambda E - A| = \begin{vmatrix} \lambda & i & -1 \\ -i & \lambda & 0 \\ -1 & 0 & \lambda \end{vmatrix} = \lambda(\lambda^2 - 2)$$

$$\lambda_1 = 0, \lambda_2 = \sqrt{2}, \lambda_3 = -\sqrt{2}$$

$$\lambda_1 = 0 \text{ 的单位特征向量 } \alpha_1 = (0, -\frac{i}{\sqrt{2}}, \frac{1}{\sqrt{2}})^T$$

$$\lambda_2 = \sqrt{2} \text{ 的单位特征向量 } \alpha_2 = (\frac{\sqrt{2}}{2}, \frac{i}{2}, \frac{1}{2})^T$$

$$\lambda_3 = -\sqrt{2} \text{ 的单位特征向量 } \alpha_3 = (-\frac{\sqrt{2}}{2}, \frac{i}{2}, \frac{1}{2})^T$$

$$\text{则 } U = (\alpha_1, \alpha_2, \alpha_3) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{i}{\sqrt{2}} & \frac{i}{2} & \frac{i}{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\text{且 } U^H A U = \begin{bmatrix} 0 & \sqrt{2} & \\ & \sqrt{2} & \\ & & -\sqrt{2} \end{bmatrix}$$

令 $X = UY$ 或 $\lambda f(x)$ 得: 其中 $Y = (y_1, y_2, y_3)^T$

$$f(x) = X^H A X$$

$$= Y^H U^H A U Y = \sqrt{2} y_2 y_2 - \sqrt{2} y_3 y_3$$